Two Isometric Classical Subspaces of Supercyclic Operators Beyond ℓ_1

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Abstract

Alves and Souza ask which Banach spaces other than ℓ_1 admit an isometric copy of a classical infinite-dimensional Banach space inside the set of supercyclic operators, apart from zero. We give two explicit positive cases. First, on c_0 the map

$$(\lambda_k)_{k\geq 1} \longmapsto \sum_{k\geq 1} \lambda_k B^k,$$

where B is the unweighted backward shift, is an isometry from ℓ_1 whose nonzero values are all supercyclic. This holds over both real and complex scalars. Second, on every infinite-dimensional separable complex Hilbert space, the supercyclic operators together with zero contain a closed isometric copy of the disk algebra $A(\mathbb{D})$. The first proof uses an exact finite-support right inverse and the Supercyclicity Criterion. The second uses adjoint Hardy multipliers and the Godefroy–Shapiro eigenvector criterion. These results answer two canonical cases of the source question; they do not classify all Banach spaces.

1 Source question and scope

The source is T. R. Alves and G. C. Souza, *On the set of supercyclic operators*, arXiv:2403.18678, later published in the *Bulletin of the London Mathematical Society* [1]. Its Theorem 1.1 proves spaceability for every separable Banach space and, more sharply, constructs an isometric copy of ℓ_1 when the underlying space is ℓ_1. Question 3.1 on PDF page 4 then asks:

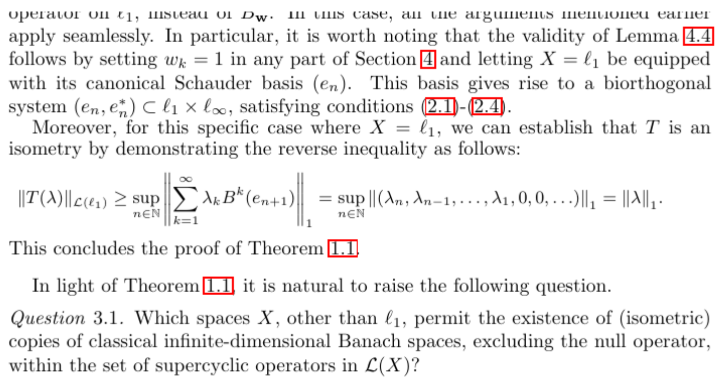
> Which spaces X, other than ℓ_1, permit the existence of (isometric) copies of classical infinite-dimensional Banach spaces, excluding the null operator, within the set of supercyclic operators in $\mathcal{L}(X)$?

See Figure 1.

The wording is classificatory. Our two theorems give complete constructions for c_0 and for separable complex Hilbert spaces, but do not characterize all X. We therefore record them as a substantial partial answer.

2 Proof intuition

For c_0, use exactly the operator series that gives the source's isometric ℓ_1 copy on ℓ_1. The upper operator-norm estimate is the triangle inequality. The c_0 norm gives the reverse estimate even more directly: one finite unit vector can align the phases of any finite initial block of coefficients in a single output coordinate.



operator on ℓ_1, instead of $B_{\mathbf{w}}$. In this case, all the arguments mentioned earlier apply seamlessly. In particular, it is worth noting that the validity of Lemma 4.4 follows by setting $w_k = 1$ in any part of Section 4 and letting $X = \ell_1$ be equipped with its canonical Schauder basis (e_n). This basis gives rise to a biorthogonal system $(e_n, e_n^*) \subset \ell_1 \times \ell_\infty$, satisfying conditions (2.1)-(2.4).

Moreover, for this specific case where $X = \ell_1$, we can establish that T is an isometry by demonstrating the reverse inequality as follows:

$$\|T(\lambda)\|_{\mathcal{L}(\ell_1)} \geq \sup_{n\in\mathbb{N}} \left\| \sum_{k=1}^{\infty} \lambda_k B^k(e_{n+1}) \right\|_1 = \sup_{n\in\mathbb{N}} \|(\lambda_n, \lambda_{n-1}, \ldots, \lambda_1, 0, 0, \ldots)\|_1 = \|\lambda\|_1.$$

This concludes the proof of Theorem 1.1.

In light of Theorem 1.1, it is natural to raise the following question.

Question 3.1. Which spaces X, other than ℓ_1, permit the existence of (isometric) copies of classical infinite-dimensional Banach spaces, excluding the null operator, within the set of supercyclic operators in $\mathcal{L}(X)$?

Figure 1: Question 3.1 and its immediate context in arXiv:2403.18678, PDF p. 4.

Supercyclicity is also transparent on finite supports. If the first nonzero coefficient is λ_p, factor the operator as $B^p(\lambda_p I + N)$, where N strictly lowers supports. Thus N is locally nilpotent and a finite Neumann series inverts $\lambda_p I + N$ on c_{00}. Composing this inverse with the forward shift produces a right inverse. Forward powers of the original operator eventually annihilate every finitely supported vector, which verifies the Supercyclicity Criterion without any growth estimate.

For Hilbert space, work on $H^2(\mathbb{D})$. Adjoint multiplication by a nonconstant analytic symbol has reproducing kernels as eigenvectors. After rescaling a nonzero symbol that vanishes at zero, its eigenvalues occur both inside and outside the unit circle, with dense spans on both sides. The Godefroy–Shapiro criterion makes the rescaled operator hypercyclic, hence the original operator supercyclic. A conjugate reflection of the symbol corrects the conjugate-linearity of taking adjoints and produces a genuinely complex-linear isometric embedding of the disk algebra.

3 An isometric ℓ_1 copy on c_0

Let $\mathbb{K}$ be $\mathbb{R}$ or $\mathbb{C}$, let $c_0 = c_0(\mathbb{N}_0; \mathbb{K})$, and define the backward and forward shifts by

$$B(x_0, x_1, x_2, \ldots) = (x_1, x_2, \ldots), \qquad F(x_0, x_1, \ldots) = (0, x_0, x_1, \ldots).$$

Write c_{00} for the finitely supported sequences.

Theorem 1 (The c_0 construction)**.** *The map*

$$\Phi : \ell_1(\mathbb{K}) \longrightarrow \mathcal{L}(c_0), \qquad \Phi(\lambda) = T_\lambda := \sum_{k=1}^{\infty} \lambda_k B^k, \tag{1}$$

is a linear isometry. Moreover, T_λ is supercyclic whenever $\lambda \neq 0$. Consequently,

$$\Phi(\ell_1) \setminus \{0\} \subseteq \mathrm{SC}(c_0),$$

and $\mathrm{SC}(c_0) \cup \{0\}$ contains a closed isometric copy of ℓ_1.

Proof. Because $\|B\| = 1$, the series in (1) converges in operator norm and

$$\|T_\lambda\| \le \sum_{k\ge 1} |\lambda_k| = \|\lambda\|_1. \tag{2}$$

For $N \ge 1$, choose $x^{(N)} \in c_{00}$ with $\|x^{(N)}\|_\infty \le 1$ and, for $j \ge 0$,

$$x_j^{(N)} = \begin{cases} \overline{\lambda_{j-1}}/|\lambda_{j-1}|, & 2 \le j \le N+1 \text{ and } \lambda_{j-1} \ne 0, \quad (\mathbb{K} = \mathbb{C}), \\ \operatorname{sgn}(\lambda_{j-1}), & 2 \le j \le N+1 \text{ and } \lambda_{j-1} \ne 0, \quad (\mathbb{K} = \mathbb{R}), \\ 0, & \text{otherwise.} \end{cases}$$

Thus $x_{k+1}^{(N)}$ cancels the phase of λ_k for $1 \le k \le N$. Hence

$$\|T_\lambda\| \ge |(T_\lambda x^{(N)})_0| = \sum_{k=1}^{N} |\lambda_k|.$$

Letting $N \to \infty$ and using (2) proves that Φ is an isometry.

It remains to prove supercyclicity. Fix $0 \ne \lambda \in \ell_1$, and put

$$p = \min\{k \ge 1 : \lambda_k \ne 0\}, \qquad A = \sum_{j=0}^{\infty} \lambda_{p+j} B^j = \lambda_p I + N,$$

where $N = \sum_{j\ge 1} \lambda_{p+j} B^j$. Then

$$T_\lambda = B^p A = A B^p. \tag{3}$$

On c_{00} the operator N strictly lowers the largest support index, so it is locally nilpotent. Therefore

$$Ry = \lambda_p^{-1} \sum_{r=0}^{\infty} (-\lambda_p^{-1} N)^r y, \qquad y \in c_{00}, \tag{4}$$

is a finite sum for each y, maps c_{00} into itself, and satisfies $AR = RA = I$ there. Define $S : c_{00} \to c_{00}$ by

$$S = RF^p.$$

Since $B^p F^p = I$, equations (3)–(4) give

$$T_\lambda S = B^p A R F^p = I \quad \text{on } c_{00}. \tag{5}$$

We now apply the Supercyclicity Criterion in the form recalled by the source [1]. Take both dense test sets to be c_{00}, take $n_j = j$, and take $S_j = S^j$. For fixed $x, y \in c_{00}$, the operator A does not increase the largest support index. By (3),

$$T_\lambda^j x = B^{pj} A^j x = 0$$

for all sufficiently large j. Consequently

$$\|T_\lambda^j x\| \, \|S^j y\| \longrightarrow 0.$$

Also, (5) and associativity give

$$T_\lambda^j S^j y = y$$

for every j. These are precisely the two clauses of the Supercyclicity Criterion, so T_λ is supercyclic. $\square$

4 The Hilbert-space construction

Let $A(\mathbb{D})$ denote the disk algebra with the supremum norm, and put

$$A_0(\mathbb{D}) = \{f \in A(\mathbb{D}) : f(0) = 0\}.$$

For $f \in A(\mathbb{D})$ define $f^{\#}(z) = \overline{f(\overline{z})}$. On the Hardy space $H^2 = H^2(\mathbb{D})$, let M_g be multiplication by g.

Lemma 2 (Inside/outside eigenvector criterion)**.** *Let T be a bounded operator on a separable complex Banach space. If the spans of the eigenvectors associated respectively with eigenvalues of modulus less than one and greater than one are both dense, then T is hypercyclic.*

Proof. This is the Godefroy–Shapiro criterion [2]. On the span of the eigenvectors with $|\lambda| < 1$, T^n tends pointwise to zero. On the span of eigenvectors with $|\mu| > 1$, define the approximate inverse at time n by multiplying a μ-eigenvector by μ^{-n}. These maps tend pointwise to zero and are exact right inverses for T^n. The Hypercyclicity Criterion applies. □

Theorem 3 (The Hilbert-space construction)**.** *Let H be an infinite-dimensional separable complex Hilbert space. Then $\mathrm{SC}(H) \cup \{0\}$ contains a closed complex-linear isometric copy of the disk algebra $A(\mathbb{D})$.*

Proof. It suffices by unitary equivalence to work on $H^2(\mathbb{D})$. Define

$$\Psi : A_0(\mathbb{D}) \longrightarrow \mathcal{L}(H^2), \qquad \Psi(f) = M^*_{f^{\#}}. \tag{6}$$

Both $f \mapsto f^{\#}$ and taking the adjoint are conjugate-linear, so Ψ is complex-linear. Moreover,

$$\|\Psi(f)\| = \|M_{f^{\#}}\| = \|f^{\#}\|_\infty = \|f\|_\infty. \tag{7}$$

For completeness, the multiplier-norm equality follows from the upper bound $\|M_g\| \le \|g\|_\infty$ and the kernel identity

$$M_g^* k_w = \overline{g(w)} k_w, \qquad k_w(z) = \frac{1}{1 - \overline{w}z}, quad w \in \mathbb{D},$$

which gives the reverse bound after taking the supremum over w.

Fix $0 \ne f \in A_0(\mathbb{D})$. Then $f^{\#}$ is nonconstant and $f^{\#}(0) = 0$. Choose $\alpha \ne 0$ so large that the analytic function

$$\psi = \overline{\alpha} f^{\#}$$

has a value of modulus greater than one in $\mathbb{D}$. The sets

$$U_- = \{w \in \mathbb{D} : |\psi(w)| < 1\}, \qquad U_+ = \{w \in \mathbb{D} : |\psi(w)| > 1\}$$

are nonempty and open: U_- contains a neighborhood of zero, while U_+ is nonempty by the choice of α. For every nonempty open $U \subseteq \mathbb{D}$, the kernels $\{k_w : w \in U\}$ have dense span in H^2; indeed, a function orthogonal to all of them vanishes on U and hence vanishes identically. Since

$$\alpha\Psi(f) = M_\psi^*, \qquad M_\psi^* k_w = \overline{\psi(w)} k_w,$$

Lemma 2 shows that $\alpha\Psi(f)$ is hypercyclic. Its orbit is contained in the projective orbit of $\Psi(f)$ because $(\alpha\Psi(f))^n = \alpha^n \Psi(f)^n$. Thus $\Psi(f)$ is supercyclic.

Finally, multiplication by z maps $A(\mathbb{D})$ onto $A_0(\mathbb{D})$ and is an isometry, because the supremum norm of every disk-algebra function is its supremum on the unit circle. Combining this isometry with (6) and (7) proves the result on H^2, and unitary conjugation transfers it to H. □

Corollary 4 (Partial answer to Question 3.1)**.** *Question 3.1 has explicit positive answers for $X = c_0$ and for every infinite-dimensional separable complex Hilbert space. The corresponding classical spaces can be taken to be ℓ_1 and $A(\mathbb{D})$, respectively.*

5 Verification, limitations, and novelty

Verification. In Theorem 1, the lower norm estimate uses only finite vectors, so no forbidden constant-modulus element of c_0 is introduced. The inverse series in (4) is finite on each input; no convergence of that series as a bounded operator is asserted or needed. The Supercyclicity Criterion does not require a uniform estimate on $S^j y$, because $T_\lambda^j x$ is eventually exactly zero.

For Theorem 3, the reflection $f^\#$ is essential: without it, $f \mapsto M_f^*$ would be conjugate-linear. The two open eigenvalue regions are nonempty by the vanishing at zero and the scalar rescaling. Density of their kernel spans follows directly from analytic uniqueness. These checks are expanded in the accompanying `VERIFICATION.md`.

Limitations. The c_0 theorem works over both scalar fields. The disk-algebra theorem is stated over the complex field, where analytic multiplier and hypercyclicity arguments apply. No claim is made for nonseparable Hilbert spaces, which cannot support a supercyclic operator, or for a classification of all separable Banach spaces.

Novelty check. Cheap run/corpus indexes and bounded searches by arXiv id, title, authors, the exact question, and combinations of "isometric copy", c_0, Hilbert space, disk algebra, and supercyclic operators found no later answer. The only indexed paper citing the source is a 2026 article on a different frequent-supercyclicity criterion [3]; it does not identify an isometric classical subspace of $\mathcal{L}(c_0)$ or $\mathcal{L}(H)$. This is a bounded search, not an exhaustive novelty guarantee. Independent mathematical and literature review is recommended before external use.

References

[1] T. R. Alves and G. C. Souza, *On the set of supercyclic operators*, Bull. Lond. Math. Soc. **56** (2024), 2483–2494; arXiv:2403.18678, `https://arxiv.org/abs/2403.18678`.

[2] G. Godefroy and J. H. Shapiro, *Operators with dense, invariant, cyclic vector manifolds*, J. Funct. Anal. **98** (1991), 229–269, `https://doi.org/10.1016/0022-1236(91)90078-J`.

[3] T. R. Alves, G. Botelho, and V. V. Fávaro, *On frequently supercyclic operators and an* $(\mathcal{F}, \Gamma)$*-supercyclicity criterion with applications*, J. Math. Anal. Appl. **556** (2026), 130158, `https://doi.org/10.1016/j.jmaa.2025.130158`.